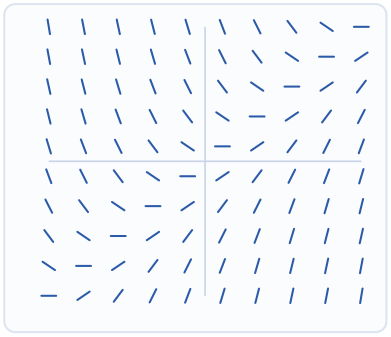
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Power Series Solutions & Method of Frobenius Practice Exercises

1 MCQ

For $(x^2 - x)y'' + y' - y = 0$, the point $x = 0$ is:



A ordinary point

B a point where no solution exists

C singular point

D a point where every solution vanishes

2 MCQ

Near $x_0 = 0$, the power series solutions of $(x + 4)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$

A 5

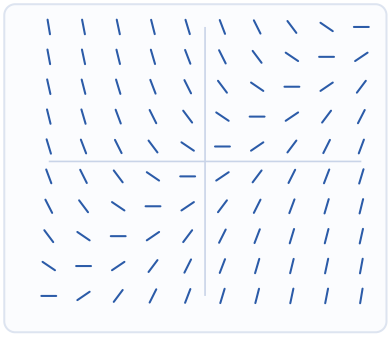
B 4

C ∞

D 1

3 MCQ

For $(x^2 - 9)y'' + y' - y = 0$, the point $x = -1$ is:



A singular point

B ordinary point

C a point where no solution exists

D a point where every solution vanishes

4 MCQ

Near $x_0 = 0$, the power series solutions of $(x^2 - 4x + 3)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$

A 3

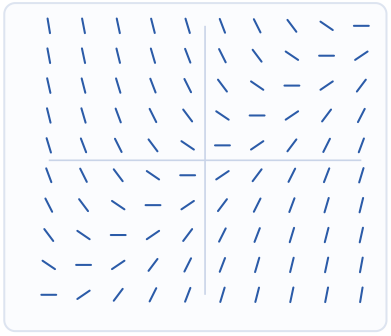
B ∞

C 1

D 2

5 MCQ

For $(x^2 + x)y'' + y' - y = 0$, the point $x = -1$ is:



A singular point

B ordinary point

C a point where no solution exists

D a point where every solution vanishes

6 MCQ

Near $x_0 = 0$, the power series solutions of $(x^2 + 4)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$

A 2

B ∞

C 1

D 4

7 MCQ

For $(x^2 - 4x)y'' + y' - y = 0$, the point $x = 0$ is:

A ordinary point

B a point where no solution exists

C a point where every solution vanishes

D singular point

8 MCQ

Near $x_0 = 0$, the power series solutions of $(x^2 + x - 6)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$

A 2

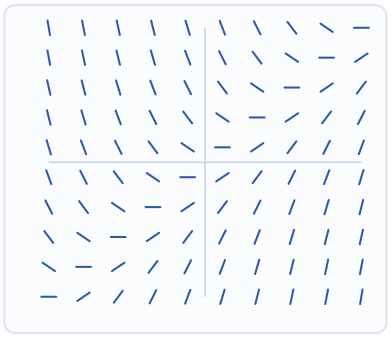
B 3

C ∞

D 1

9 MCQ

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients c_n.



A $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

B $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

C $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

D $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

10 MCQ

The Laplace transform of $h(u) = 190e^{20u}$ is $k(s) = \dfrac{190}{s-20}$ for $s > 20$. Determine the evaluated transform $k(21)$.

A 190

B 209

C 170

D 193

11 MCQ

Consider the second-order linear homogeneous differential equation: $x^2y'' + xy' + (x^2 - \nu^2)y = 0$, where ν is a constant. This is Bessel's equation of order ν. Assume $\nu = 1/2$ and that $x > 0$.

Part 1: Ordinary Point Analysis

$x = 0$ is a regular singular point. Determine the indicial equation for this differential equation and find its roots.

A Indicial equation: $r^2 + \frac{1}{4} = 0$; roots: $r = \pm\frac{i}{2}$

B Indicial equation: $r^2 - r + \frac{1}{4} = 0$; roots: $r = \frac{1}{2}, \frac{1}{2}$

C Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{4}$

D Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{2}$

12 MCQ

Consider the differential equation $y'' - xy = 0$. We are looking for a power series solution of the form $y(x) = \sum_{n=0}^{\infty} a_n x^n$ around the ordinary point $x = 0$. Find the coefficient a_2 in terms of a_0 and a_1.

A $a_2 = a_0 + a_1$

B $a_2 = 0$

C $a_2 = a_0$

D $a_2 = a_1$

13 MCQ

The first-order differential equation $\frac{du}{ds} = 17s - 6u$ has its slope field plotted below. Determine the slope $\frac{du}{ds}$ at the point $K(14, 7)$.

A 189

B 213

C 196

D 199

14 MCQ

For $(x^2 + x)y'' + y' - y = 0$, the point $x = -2$ is:

A singular point

B ordinary point

C a point where no solution exists

D a point where every solution vanishes

15 MCQ

Near $x_0 = 0$, the power series solutions of $(x^2 + 6)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$

A ∞

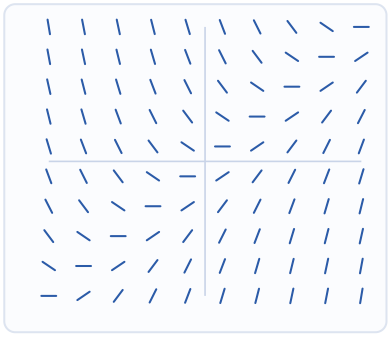
B 1

C $2\sqrt{6}$

D $\sqrt{6}$

16 MCQ

For $(x^2 - 16)y'' + y' - y = 0$, the point $x = 4$ is:



A singular point

B a point where no solution exists

C a point where every solution vanishes

D ordinary point

17 MCQ

For $(x^2 + 2x)y'' + y' - y = 0$, the point $x = 3$ is:

A a point where no solution exists

B a point where every solution vanishes

C singular point

D ordinary point

18 MCQ

For $(x^2 - x)y'' + y' - y = 0$, the point $x = 2$ is:

- **A** ordinary point
- **B** singular point
- **C** a point where no solution exists
- **D** a point where every solution vanishes

19 MCQ

Near $x_0 = 0$, the power series solutions of $(x^2 + 5)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$

- **A** ∞
- **B** 1
- **C** $\sqrt{5}$
- **D** $2\sqrt{5}$

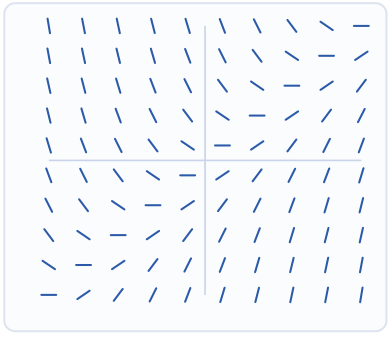
20 MCQ

For $(x^2 - 16)y'' + y' - y = 0$, the point $x = -1$ is:

- **A** singular point
- **B** ordinary point
- **C** a point where no solution exists
- **D** a point where every solution vanishes

21 MCQ

Near $x_0 = 1$, the power series solutions of $(x - 3)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$



- **A** 1
- **B** 3
- **C** 2
- **D** ∞

22 MCQ

For $(x^2 - 1)y'' + y' - y = 0$, the point $x = -1$ is:

- **A** ordinary point
- **B** a point where no solution exists
- **C** singular point
- **D** a point where every solution vanishes

23 MCQ

For $(x^2-1)y'' + y' - y = 0$, the point $x = 1$ is:

- **A** singular point
- **B** ordinary point
- **C** a point where no solution exists
- **D** a point where every solution vanishes

24 MCQ

Near $x_0 = 1$, the power series solutions of $(x+3)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$

- **A** 4
- **B** 5
- **C** ∞
- **D** 1

25 MCQ

Near $x_0 = 0$, the power series solutions of $(x+2)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$

- **A** 3
- **B** ∞
- **C** 1
- **D** 2